Late Deliveries and Customer Satisfaction – Olist (Power BI)

Power BI dashboard that answers one question: **do late deliveries hurt customer satisfaction, and if they do, where and why do they happen?**

Course project – Business Intelligence (M1, UPEC) – Bernardo Gomes

Power BI file and PDF: Google Drive folder

The case

Olist is a Brazilian marketplace that connects small shops to large online stores. The public dataset contains about 99,000 real, anonymized orders placed between 2016 and 2018, split into 9 CSV tables.

- Dataset: Brazilian E-Commerce Public Dataset by Olist (Kaggle)
- Period analyzed: January 2017 to August 2018 (96,203 delivered orders)

Main results

Finding	Value
Share of orders delivered late	6.8%
Average review, on-time vs late orders	4.3 vs 2.3
Worst month	March 2018 (19% late)
Worst customer state	Alagoas (21% late)
Days with the carrier, late vs on-time orders	27 vs 7.5
Late rate when the seller ships after the deadline	21% (vs 5%)
Late orders that started with a late shipment by the seller	28%

Dashboard pages

1. **Overview** – the problem: late rate over time and its impact on the review score.
2. **Where** – late rate by customer state, seller state, product category and top sellers.
3. **Why and what to do** – causes (carrier time, late shipping by sellers) and four recommendations.

How it was built (summary, full details in BUILD_GUIDE.md)

1. Imported 7 of the 9 CSV files into Power BI Desktop (Get data → Text/CSV, locale English US).
2. Power Query:
   - Orders: kept only delivered orders with a delivery date; added `DelayDays`, `DeliveryStatus` (Late / On time) and `DeliveryDays`.
   - Reviews: grouped by order and averaged the score (some orders had several reviews).
   - Category translation: promoted the first row to headers.
3. Model: star schema with **Orders** in the center, many-to-one relationships, single filter direction.
4. DAX measures: Total Orders, Late Orders, Late %, Avg Review, Avg Delivery Days, and " (Items)" versions for the seller and category analysis (using `CROSSFILTER`).
5. DAX columns: Purchase Month, Seller, CarrierDays, SellerShipping.
6. Report filter: purchases from January 2017 (2016 had too few orders).

Files

File	Content
`INSTALL.md`	How to install Power BI Desktop and get the data
`BUILD_GUIDE.md`	Step-by-step guide to rebuild the dashboard (Power Query, model, DAX, pages)
`Olist_Dashboard.pdf`	The three dashboard pages as PDF
`README.md`	This page

The Power BI file (`Projeto_BI_Olist_Atrasos.pbix`) is larger than GitHub's 25 MB upload limit, so it is shared on Google Drive: download the .pbix file.

The CSV files are not included: download them from the Kaggle link above.

How to open the project

1. Install Power BI Desktop (see INSTALL.md).
2. Download the `.pbix` file from the Google Drive folder and open it.
3. To refresh the data from your own copy of the CSVs: Home → Transform data → Data source settings → Change Source, and point each table to your folder.